Next Materials 4 (2024) 100114

Contents lists available at ScienceDirect

Next Materials

journal homepage: www.sciencedirect.com/journal/next-materials

Research Article

A density functional theory research on the interaction of H_2X molecules with the Al_2C nanosheet (X = O, S, Se, Te)

Nahuel Moreno Yalet [a], Víctor A. Ranea [a,b,*]

[a] CCT-CONICET-La Plata. Instituto de Investigaciones Fisicoquímicas Teóricas y Aplicadas (INIFTA), Facultad de Ciencias Exactas, Universidad Nacional de La Plata, Calle 64 y Diagonal 113, 1900 La Plata, Argentina

[b] Departamento de Física, Facultad de Ciencias Exactas, Universidad Nacional de La Plata, Calle 49 y Calle 115, 1900 La Plata, Argentina

ARTICLE INFO

Keywords:
Al2C nanosheet
Hazardous small molecules
Water molecule
Adsorption
Dissociation
Sensor
Density functional theory
Catalysis

ABSTRACT

The potential application of the 2D material Al_2C, as catalysts and as sensor of the hazardous molecules H_2S, H_2Se and H_2Te, is studied using theoretical methods. Density functional theory (DFT) has been applied to study the interaction between the pristine Al_2C nanosheet and small molecules, such as H_2O, H_2S, H_2Se and H_2Te. Results show the Al_2C nanosheet is a promising material to dissociate the studied molecules as well as its potential use as sensor of these molecules in molecular or dissociated state. Molecular interactions are weak even considering the London dispersion forces. Partial dissociative adsorption is strongly preferred to molecular adsorption for all four molecules. Applying the climbing image NEB method, the activation energy for H_2O dissociation to OH-H is calculated in $\approx$ 0.1 eV whereas it is just a few meV for H_2S, H_2Se and H_2Te partial dissociation. The calculated energy gap for the clean Al_2C nanosheet is 1.078 eV whereas it is increased to 1.371, 1.338, 1.379 and 1.416 eV for molecular adsorption and to 1.209, 1.363, 1.370 and 1.388 eV for partial dissociative adsorption.

1. Introduction

The carbon atom is known for its ability to form hibridized orbitals, in molecules as well as in solid state. In 2D graphyne layers, the carbon atom exhibits *sp* and sp^2 hibridization [1,2]. In graphene (2D) [3], carbon nanotubes 1D [4,5] and fullerene [6] the hibridization exhibited is sp^2 whereas in diamond (3D), it is sp^3. Beyond this richness, several years ago, the possible existence of planar tetracoordinate carbon (ptC) was predicted [7–9] and a few years ago, the Al_2C nanosheet or monolayer was predicted using density functional theory [10,11]. According to previous calculations, the Al_2C monolayer is semiconducting with a reported energy gap of 1.05 [10] and 1.02 eV [11] using the HSE06 calculations. Such semiconductors band gap offer promising electronic and optoelectronic applications. The computed cohesive energy suggests the Al_2C nanosheet is a moderated 2D structure. Although the cohesive energies are not comparable directly, for the Al_2C monolayer the cohesive energy is 4.49 eV/atom whereas 7.95 and 3.51 eV/atom are the cohesive energies for graphene and bulk Al, respectively, given as reference values [11]. The kinetic stability of the Al_2C monolayer was established after the computed phonon spectrum showed no appreciable imaginary modes [11]. The thermal stability of the Al_2C monolayer at 1500 K was established with ab initio molecular dynamics simulations. Although pronounced in-plane and out of the plane distortions were found at this temperature, the structure is intact whereas at 2000 K serious deformations were found suggesting the melting temperature is between 1500 and 2000 K [11]. Although there are no experimental reports about the Al_2C nanosheet, to our knowledge, a previous article found that the calculated lattice constants of this 2D material are nearly the same as those of the PdO-terminated PdO (100) surface. It was suggested that the PdO(100) surface could be used as a template to synthesize the Al_2C monolayer [10].

Small molecules, such as H_2O, H_2S, H_2Se and H_2Te, have great impact on several topics of study and on life in our planet. These four mentioned molecules have similar electronic structure. However, the H_2O molecule is required for the human life whereas the other three molecules are hazardous to humans. It is known that hydrogen sulfide (H_2S) is a colorless gas with odor of rotten egg, very toxic, corrosive and flammable. It is produced in swamps, sewers and also in volcanos, among other sources. Hydrogen selenide (H_2Se) is a very toxic (more toxic than H_2S), colorless and flammable gas. Hydrogen telluride is also

* Corresponding author at: CCT-CONICET-La Plata. Instituto de Investigaciones Fisicoquímicas Teóricas y Aplicadas (INIFTA), Facultad de Ciencias Exactas, Universidad Nacional de La Plata, Calle 64 y Diagonal 113, 1900 La Plata, Argentina.

E-mail address: vranea@inifta.unlp.edu.ar (V.A. Ranea).

https://doi.org/10.1016/j.nxmate.2024.100114
Received 1 March 2023; Received in revised form 18 December 2023; Accepted 5 January 2024
Available online 28 March 2024
2949-8228/

a colorless gas, unstable in air but can be detected by the odor of rotting garlic.

The detection of toxic molecules in the air, in the atmosphere, is essential to human wellbeing. Hazardous gases detection is important for controlling emissions resulting from human activities, such as industrial activities and the use of vehicles. That is why it is of great importance that sensors work well at room and high temperature, as generally those based in semiconductors materials do. In the last years several devices for detecting CO_2, CO, SO_2, NH_3, etc, have been developed [12]. Only in recent years, the adsorption and detection performance of the Al_2C monolayer has been investigated for some organic molecules [13] and for small molecules with XH_3 structure, with X = N, P, As and Sb [14]. In all cases, the Al_2C monolayer keeps the semiconductor structure upon adsorption changing the energy gap. The adsorption configurations of these four molecules display similar structures, adsorption via the X atom to an Al atom of the monolayer, with adsorption energies in the range − 0.43 to − 1.19 eV. In the cases of NH_3, PH_3 and SbH_3 adsorption, the energy gap increases whereas for AsH_3 adsorption the energy gap decreases strongly [14]. The interaction between the organic molecules and the Al_2C monolayer prove to be stronger with adsorption energies in the range − 1.972 to − 2.244 eV/molecule. The energy gap of system organic adsorbate-Al_2C increases from 0.9 eV to 1.41 and 1.57 after interaction with ethylene oxide and vinyl chloride; whereas decreases to 0.52 and 0.42 eV upon acetaldehyde and benzene adsorption. In the present effort the potential use of the pristine Al_2C monolayer as a sensor of the molecules H_2O, H_2S, H_2Se and H_2Te is evaluated using density of states (DOS). The calculated energy gap of the pristine Al_2C layer is compared with the energy gap after molecular adsorption (H_2X).

The use of a surface to facilitate a chemical reaction or the dissociation of a specific molecule is not new. Several industrial processes are easier with the help of a catalyst. Many different catalysts are used in a large number of chemical reactions in a variety of tasks. In the present manuscript, the ability of the pristine Al_2C nanosheet to dissociate partially the H_2O, H_2S, H_2Se and H_2Te molecules is evaluated in the low coverage regime. After the study of molecular adsorption, dissociative adsorption is studied and the climbing nudged elastic band (ciNEB) methodologies are applied to calculate activation energies for dissociation after molecular adsorption.

Today, it is not possible to resolve the internal structure of a small molecule adsorbed on a surface applying experimental techniques. Density functional theory (DFT) is an advanced reliable theoretical technique useful to calculate adsorption energies, adsorption configurations, charge configurations and can be applied to describe the bonds among small molecules and a surface. The results of the present work could be useful as a step fordward in the study of the potential use of the Al_2C nanosheet as sensor for the small molecules H_2O, H_2S, H_2Se and H_2Te as well as its ability to dissociate these molecules. The inclusion of the London dispersion interactions in the calculations is an improvement in the theoretical description of the systems under consideration.

2. Methodology

Throughout this project, total energy calculations, using density functional theory, were performed as implemented in the Vienna Ab initio Simulation Package (VASP) [15,16]. The Kohn-Sham equations were solved applying the projector augmented wave (PAW) approach for describing electronic core states [17,18] and a plane-wave basis set including plane waves up to 400 eV was used whereas electron correlation and exchange energies were calculated within the generalized gradient approximation (GGA) in the Perdew-Burke-Ernzerhof (PBE) form [19,20]. This functional fails to describe the van der Waals (vdW) interactions resulting from dynamical correlations between fluctuating charge distributions (London dispersion interactions). The vdW-DF methods are non-local methods where the correlation energy is calculated using a double space integral of electron densities and an integration kernel (see [21] and references therein). The original vdW-DF tends to overestimate long range interactions, vdW-DF2 was later developed. These type of functionals have been applied to layered materials [32–34]. In them the bonds within every layer are strong (such as covalent or ionic bonds) whereas the bonds between layers are weak and are considered due to vdW interactions. One limitation to the vdW method used in this manuscript is that it considers the dispersion correction to be pairwise additive. The interaction energy between two atoms is calculated regardless of the atoms between them. This pairwise approximation does not take into account that the interaction could be screened [21]. The present manuscript includes calculations using the rPW86 (vdW-DF2) functional [26–29,30,31] adsorption energies. Whereas the PBE functional underestimates the energy gap value calculated for semiconductors, in recent studies it was found that the HSE06 hybrid functional [22] shows an accurate description of the electronic properties of semiconductors 2D materials [23,24]. In the present manuscript the HSE06 hybrid functional is applied to calculate the density of states (DOS) and the energy gap between the valence and conduction bands. The nudged elastic band (NEB) method is used to calculate the minimum energy pathways (MEP) and the climbing image NEB method (ciNEB) is applied to calculated the transition states for H_2X dissociation to HX-X with 8 images [25]. Following the construction of an initial MEP between molecular and dissociative configurations, an initial calculation is performed in order to locate the MEP. After this initial step, a ciNEB calculation was used to force the image with the highest energy to the transition state, which was converged to 0.05 eV/Å. The NEB calculations were performed using the PBE functional. After convergence, the Hessian matrix of second derivatives was calculated for all transition state structures within the harmonic approximation by two-sided finite differences, using a displacement step of 0.01 Å. Diagonalization of the dynamical matrix yields the harmonic frequencies and one imaginary frequency was found in every calculation.

An orthorhombic supercell with lattice constants [10,11]: $a = 6.019$ (x direction), $b = 10.14$ (y direction) and $c = 15$ (z direction) Å was used and periodic boundary conditions were applied in all three perpendicular directions. The surface was modelled by a monolayer slab separated by more than a 12 Å vacuum region to avoid interactions between slabs. A (2 × 2) surface was used to minimize interactions between adsorbates and all the atoms of the monolayer and of the adsorbate were allowed to move according to the calculated forces. Atomic relaxations were considered converged when the forces on the ions were less than 0.01 eV/Å. The first Brillouin zone of the supercell was sampled with a (5 × 3 × 1) Γ centred mesh, whereas a (10 × 6 × 1) Γ centred mesh was used in the DOS calculations.

The adsorption energy of the adsorbed molecules, E_a, was defined as:

$$E_a = TE(adsorbate/monolayer) - TE(adsorbate) - TE(monolayer) \quad (1)$$

The terms *adsorbate* and *monolayer* in equation (1) refers to the adsorbed species (H_2O, H_2S, H_2Se, H_2Te) and to the Al_2C nanosheet, respectively. The first term of the right hand side of the equation (1) represents the calculated total energy, *TE*, of the optimized configuration of adsorbed species on the relaxed Al_2C nanosheet; the second term stands for the *TE* of the adsorbate calculated in a 15 Å cubic supercell whereas the third term represents the *TE* of the clean optimized monolayer. With this definition, negative values of E_a stand for stable configurations.

Charge density calculations were performed for the systems *adsorbate/surface*, *adsorbate* and *surface*. First, the coordinates of the *adsorbate* and the *surface* were kept fixed after optimization of the system *adsorbate/surface*. Only then, the difference of the charge densities was calculated for every system.

3. Results and discussions

The pristine Al_2C nanosheet. The structure of the Al_2C nanosheet has been investigated using theoretical techniques only, as described in previous works [10,11]. Such research reveals that the lattice parameters are $a = 3.03$ Å and $b = 5.06$ Å according to reference [10], whereas they are $a = 3.04$ Å and $b = 5.06$ Å following reference [11] and also reference [13]. The calculated PBE results in the present manuscript are in good agreement with them: $a = 3.009$ Å and $b = 5.070$ Å, see Fig. 1. Results reveal that the optimized bond lengths between Al atoms and between Al and C atoms are 2.614 and 1.942 Å. Comparisons show that these results are in good agreement with previous reports: 2.60 and 1.96 Å [10,11], 2.57 and 1.91 Å [13]. Within the 2×2 model used in the present manuscript, the pristine Al_2C monolayer seems to be flat. This is no longer true after interaction with any of the adsorbates studied in the present manuscript.

Isolated adsorbates. The optimized structures of the isolated adsorbates (H_2O, H_2S, H_2Se and H_2Te) are not strongly modified after adsorption on the Al_2C nanosheet, due to a rather weak interaction with the Al_2C monolayer. However, the H_2Se and H_2Te molecules are easily partially dissociated to HX-H, X = Se and Te. Table 1 shows the internal structure of the molecules after adsorption on the Al_2C monolayer. In most of the cases, the H-X bonds are slightly longer after adsorption. PBE results reveal that the structure of the isolates molecules are: for H_2O: O-H= 0.973 Å and H-O-H= 104°; for H_2S: S-H= 1.349 Å and H-S-H= 92°; for H_2Se: Se-H= 1.481 Å and H-Se-H= 90°; for H_2Te: Te-H= 1.674 Å and H-Te-H= 90°.

Molecular adsorption, configurations. Many initial configurations were tested for the adsorption of every adsorbate on the Al_2C nanosheet. Comparisons of the results reveal that the final optimized configuration in every case shows similar properties: the adsorption is via the central atom (O, S, Se, Te) to an Al atom (the metal atom) of the nanosheet with the molecular plane nearly parallel to the nanosheet, as shown schematically on Fig. 2. Similar adsorption configurations have already been found on the $V_2O_5(001)$ surface although the interaction is a bit weaker with the vanadium oxide [35,36].

Although the interaction between the H_2O molecule and the Al_2C nanosheet is rather weak, this is the strongest interaction among the four systems studied in the present manuscript. The calculated adsorption energy of the H_2O molecule on the Al_2C monolayer is − 0.510 eV/H_2O. The inclusion of the van der Waals (London dispersion) interactions increases the adsorption energy to − 0.588 eV/H_2O. Results reveal the bond length between the O and Al atoms is 2.105 Å whereas the molecular structure is slightly modified as Table 1 shows. Comparisons with previous studies show that the adsorption of a H_2O molecule on top of an Al atom belonging to the Al{100} or to the Al{111} surfaces is weaker than the adsorption on the Al_2C monolayer. Using the PBE functional, the adsorption energies are − 0.350 and − 0.26 eV on the Al{100} [37] and on the Al{111} [38] surfaces, respectively. On the other hand, the interaction is stronger when the Al atom belongs to the $Al_2O_3(0001)$ metal oxide surface. The adsorption energy is − 1.15 eV/molecule for molecular H_2O [39] whereas the O-Al bond lenght (1.965 Å) is shorter than in the previous mentioned semiconductor (Al_2C nanosheet) and metal cases. Comparisons show that in all four cases the top Al atom is attracted to the H_2O molecule from the relaxed clean surface or monolayer. The inclusion of the van der Waals interactions slightly change the H_2O structure to 106°, the H-O-H angle, and to 0.987 and 0.979 Å, the OH bond lengths, after interaction with the Al_2C monolayer, whereas the O-Al bond distance is a bit longer, 2.163 Å.

Comparison with the previous case, H_2O adsorption, shows the interaction between the H_2S molecule and the Al_2C nanosheet is much weaker. The calculated adsorption energy is − 0.158 eV (− 0.263 eV/H_2S including London dispersion forces) whereas the bond distance is much longer, 2.800 Å. (3.340 Å including vdW interactions).

Results reveal that the interaction of H_2Se and H_2Te molecules with the Al_2C nanolayer is much weaker than the interaction of the H_2O molecule with the nanolayer but it is stronger than the interaction of the H_2S with the monolayer. Table 1 shows the adsorption energies are − 0.221 and − 0.296 eV/molecule calculated for H_2Se and H_2Te, respectively, whereas in both adsorption configurations the adsorbates are further from the nanosheet than in the previous cases. As displayed on Table 1, the H-X bonds are 2.817 and 2.990 Å for X = Se and Te, respectively.

Although the inclusion of the van der Waals interactions has small effect on the internal structure of the adsorbed H_2S, H_2Se and H_2Te molecules, the adsorption energies are increased by 66%, 96% and 76%, with respect to those calculated with the PBE functional, see Table 1. Results also reveal that the Al-X bond distances are longer (19%, 12% and 16% for X = S, Se and Te) than those calculated using PBE, see Table 1.

Table 1
Calculated adsorption energies (eV/molecule) of molecular and partially dissociated H_2O, H_2S, H_2Se and H_2Te molecules on the Al_2C nanosheet. Also shown, the Al-X atom bond distance (X = O, S, Se and Te), the molecular and dissociated structures upon adsorption. Distances are in Å, calculations are performed with PBE and, between parenthesis, vdW (rPW86) functionals.

	E_a PBE (vdW)	Al-X PBE (vdW)	H-X PBE (vdW)	H-X-H PBE (vdW)
H_2O	− 0.510 (−0.588)	2.105 (2.163)	1.005 0.979 (0.987 0.979)	105° (106°)
H_2S	− 0.158 (−0.263)	2.800 (3.340)	1.385 1.356 (1.356 1.353)	91° (92°)
H_2Se	− 0.221 (−0.434)	2.817 (3.169)	1.502 1.474 (1.504 1.490)	90° (91°)
H_2Te	− 0.296 (−0.520)	2.990 (3.465)	1.712 1.678 (1.709 1.685)	88° (89°)
	E_a PBE (vdW)	**Al-X PBE (vdW)**	**X-H PBE (vdW)**	**C-H_d PBE (vdW)**
HO-H_d	− 1.422 (−1.662)	1.761 (1.767)	0.970 (0.970)	1.117 (1.111)
HS-H_d	− 1.699 (−1.495)	2.457 (2.504)	1.360 (1.358)	1.116 (1.110)
HSe-H_d	− 1.869 (−1.716)	2.604 (2.673)	1.490 (1.497)	1.116 (1.110)
HTe-H_d	− 1.902 (−1.698)	2.827 (2.912)	1.682 (1.692)	1.115 (1.110)

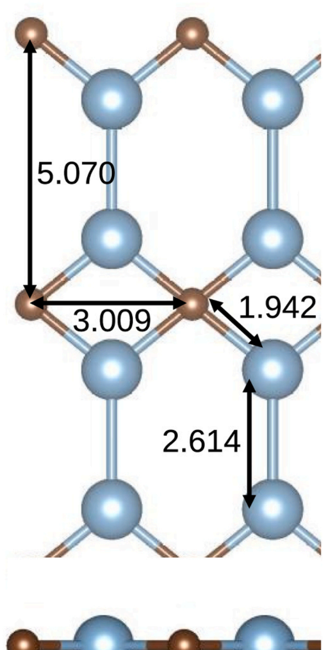

Fig. 1. Top and side views of a sphere model for the pristine Al_2C monolayer used in this manuscript. Lattice parameters a and b of the surface cell, are shown. Al and C atoms are represented by light blue and brown spheres, respectively (same in subsequent Figures).

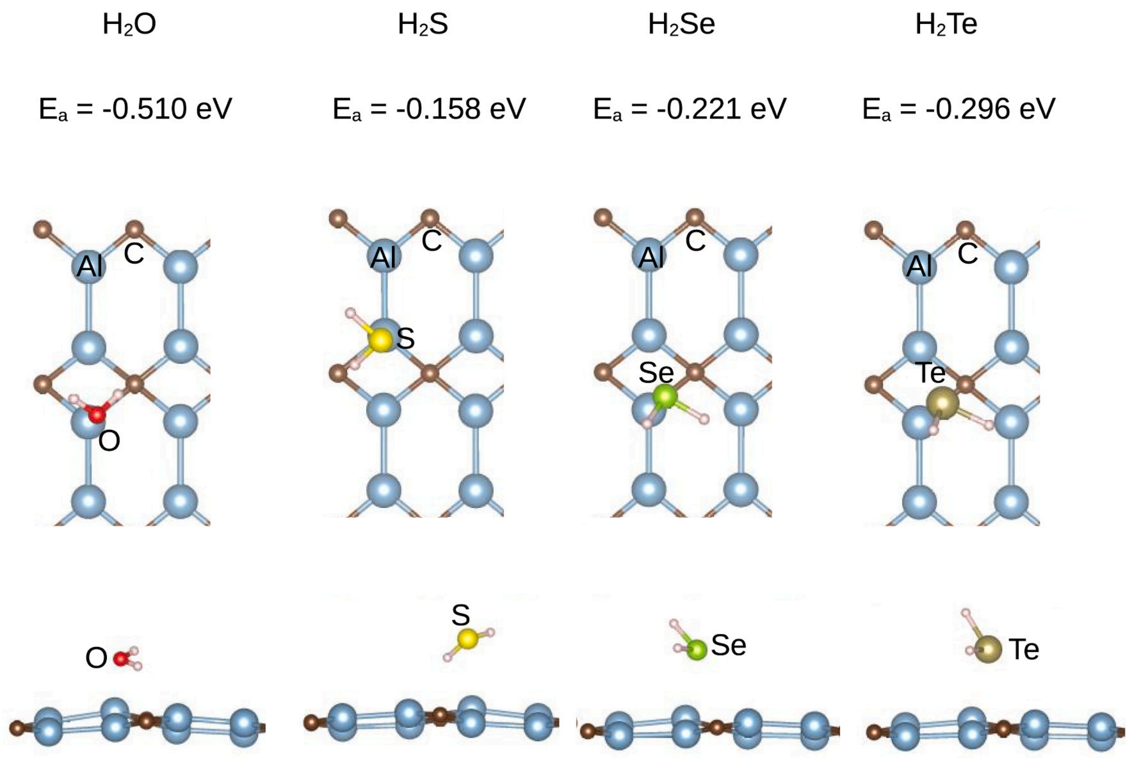

Fig. 2. Top and side views of a sphere model for H_2O, H_2S, H_2Se and H_2Te molecular adsorption on the Al_2C nanosheet. Red, yellow, light green, dark green and pink spheres stand for O, S, Se, Te and H atoms (same in subsequent Figures). PBE adsorption energies are included for every adsorbate.

Even when the interaction between any of the studied molecules and the Al_2C nanosheet is rather weak, Fig. 3 displays, schematically, that the Al_2C nanosheet seems to show roughness (distortion) after adsorption of the molecules under study (H_2O, H_2S, H_2Se and H_2Te), see also Fig. 2 bottom panels, in comparison with the flat pristine nanosheet, see Fig. 1 bottom panel. This is, probably, a consequence of the lack of support.

Molecular adsorption, charge density. Isosurfaces of difference of

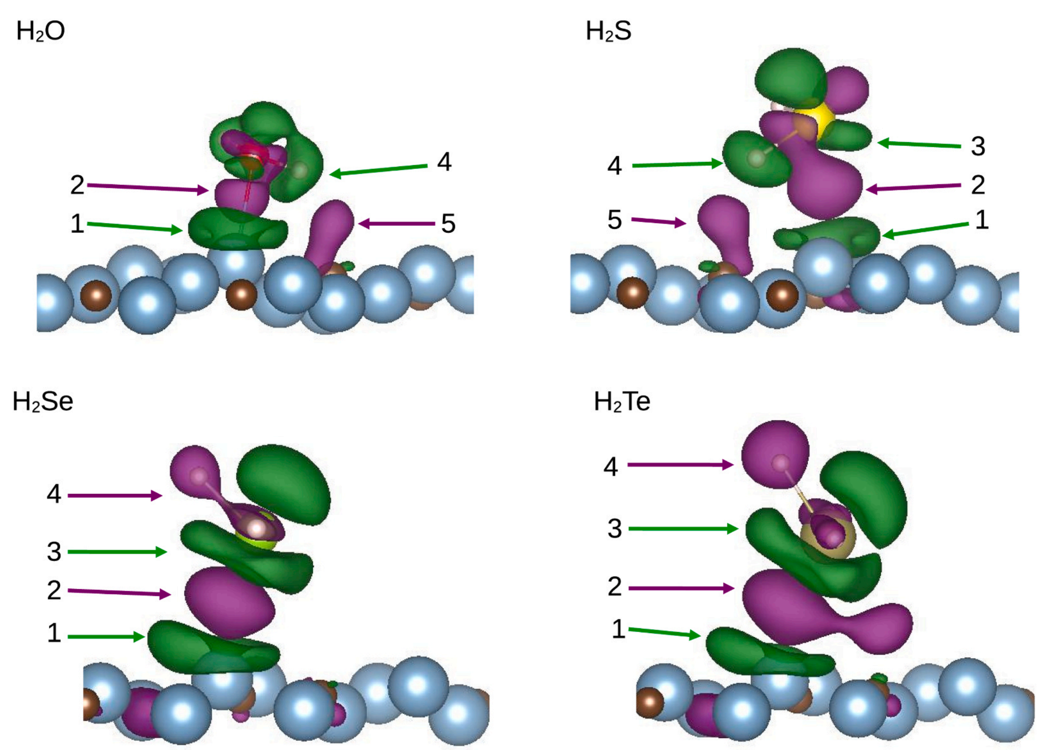

Fig. 3. Charge density difference for H_2O, H_2S, H_2Se and H_2Te molecular adsorption on the Al_2C nanosheet. The purple and green surfaces are isosurfaces where the negative charge density increases and decrease by 0.0030, 0.0015, 0.0010 and 0.0010 e/Bohr^3 for H_2O, H_2S, H_2Se and H_2Te, respectively, PBE results.

density of negative charge show a decrease of the negative charge density on the Al atom near the adsorbate in all four cases, marked with 1 on Fig. 3. On top of this region, there is an accumulation of negative charge, marked with 2 in the four cases, in one of the lone pairs of the adsorbed molecule. These two regions are similar in all four configurations.

Comparisons reveal a few differences in the charge distribution between the H_2O and H_2S molecules, on one side, and the H_2Se and H_2Te molecules, on the other. In the first case, one of the H atoms is closer to the monolayer than the central atom (O or S) whereas in the second case, the cental atom (Se or Te) is closer to the monolayer than both H atoms. Only on the biggest two molecules, there is depletion of negative charge, marked with 3 on top of the region 2. However, on the H_2S molecule seems to be a region of negative charge depletion similar to those on the H_2Se and H_2Te adsorbed molecules, also marked with 3 but it is not between the region 2 and the central atom, S, of the molecule. The H_2S molecule is smaller than the H_2Se and H_2Te molecules, the H atoms are closer to the monolayer and seem to be attracted strongly leaving the region 3 further from the monolayer.

It seems that the size of the molecules and the strong attraction between the H atoms of the adsorbates and the C atoms of the Al_2C nanosheet is related with the charge distribution on the H atoms. In both smallest molecules in which there is one H atom closer to the monolayer than the central atom, there are regions that show a decreases of the negative charge density, marked with 4 on both top panels on Fig. 3. However, on both biggest molecules there is accumulation of negative charge on the H atoms, also marked with 4 on the bottom panels of Fig. 3.

Results reveal an increase of negative charge density on C atoms near H atoms for H_2O and H_2S adsorption, marked with 5 on both top panels on Fig. 3. The size of these two molecules seems to generate the H atoms are near the C atoms to display both regions 4 and 5. However, the strongest interaction is via one lone pair of the adsorbate.

Although the numerical values given to the isosurfaces shown on Fig. 3 correspond to the PBE functional, the locations of increase and decrease of the charge density difference do not depend on the functional used in the present manuscript.

Dissociative adsorption, configurations. Results reveal that partial dissociative adsorption, XH-H (X = O, S, Se and Te), is strongly preferred energetically to molecular adsorption on the Al_2C nanosheet, as displayed on Table 1 column adsorption energy, E_a, and see Fig. 4. Comparisons among adsorption energies seem to show that the larger the XH species is, the stronger the interaction with the Al_2C monolayer is. Comparisons, also, show that the OH species is adsorbed on top of an Al atom whereas the HX (X = S, Se and Te) species are adsorbed on a bridge configuration between Al atoms with the H atom far away from the nearest C atom, see Fig. 4. In all cases, the dissociated H atom is bound on top to a C atom. Table 1 also shows the X-Al bond distances and the X-H bond lengths are longer with the size of the X atom (X = O, S, Se and Te). Same column shows the HX fragment after dissociation is closer to the nanosheet than the H_2X molecule whereas the next column shows the H-X bond length is not strongly modified after dissociation. As can be seen, schematically on Fig. 4 bottom panels, the nanosheet is strongly modified after dissociative adsorption.

Molecular to dissociative adsorption, minimum energy paths. Dissociative reactions, H_2X to HX-H for X = O, S, Se and Te, show very small calculated activation energies. Fig. 5 displays, schematically, the MEP for H_2O to HO-H reaction on the Al_2C monolayer and a sphere model for a few intermediate configurations in which the molecular adsorption energy is taken as 0.0 eV. The calculated activation energy is ≈ 0.1 eV whereas the OH bond length is stretched to 1.187 Å, see Fig. 5*c*. As can be seen, schematically in Fig. 5*d*, *e*, *f* and *g*, the Al and C atoms are displaced from the monolayer toward the adsorbates, after interaction with them (OH and H species). The O-Al bond distance is shorter after dissociation, as can be seen on Table 1, from 2.105 to 1.761 Å. On the other hand, the calculated activation energy for dissociation is about a

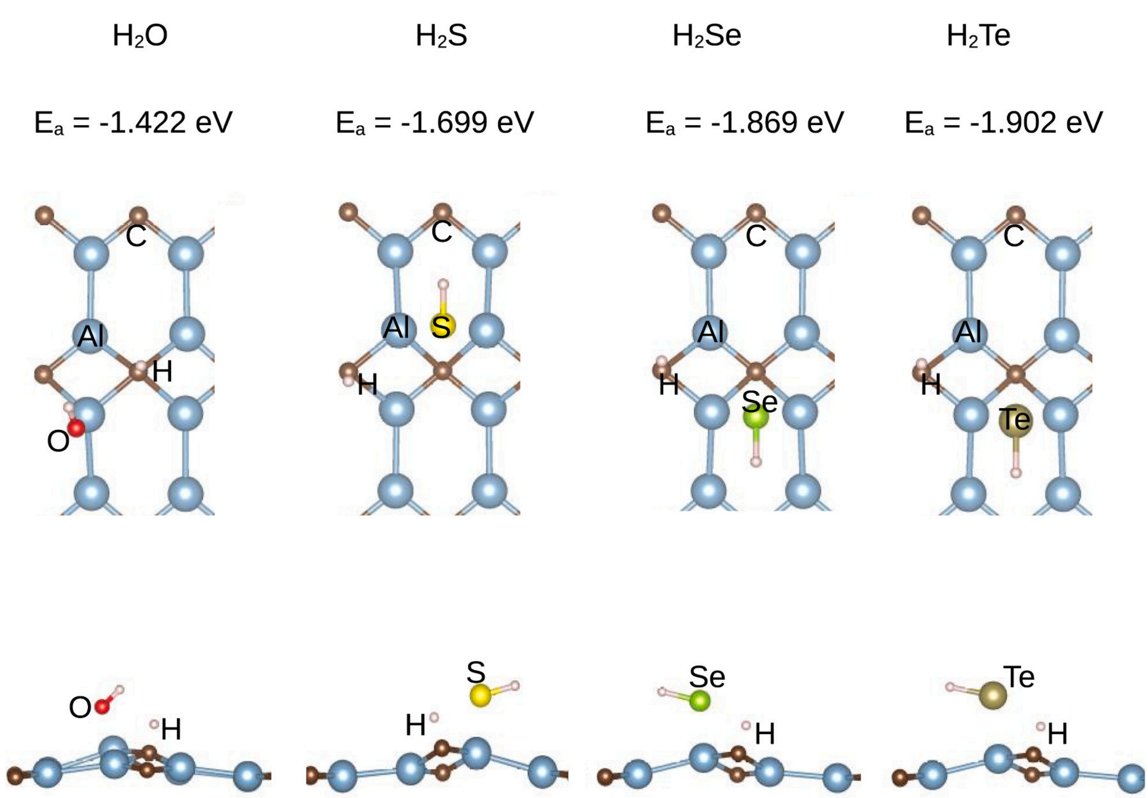

Fig. 4. Top and side views of a sphere model for H_2O, H_2S, H_2Se and H_2Te partial dissociative adsorption on the Al_2C nanosheet. PBE adsorption energies are included, see Table 2.

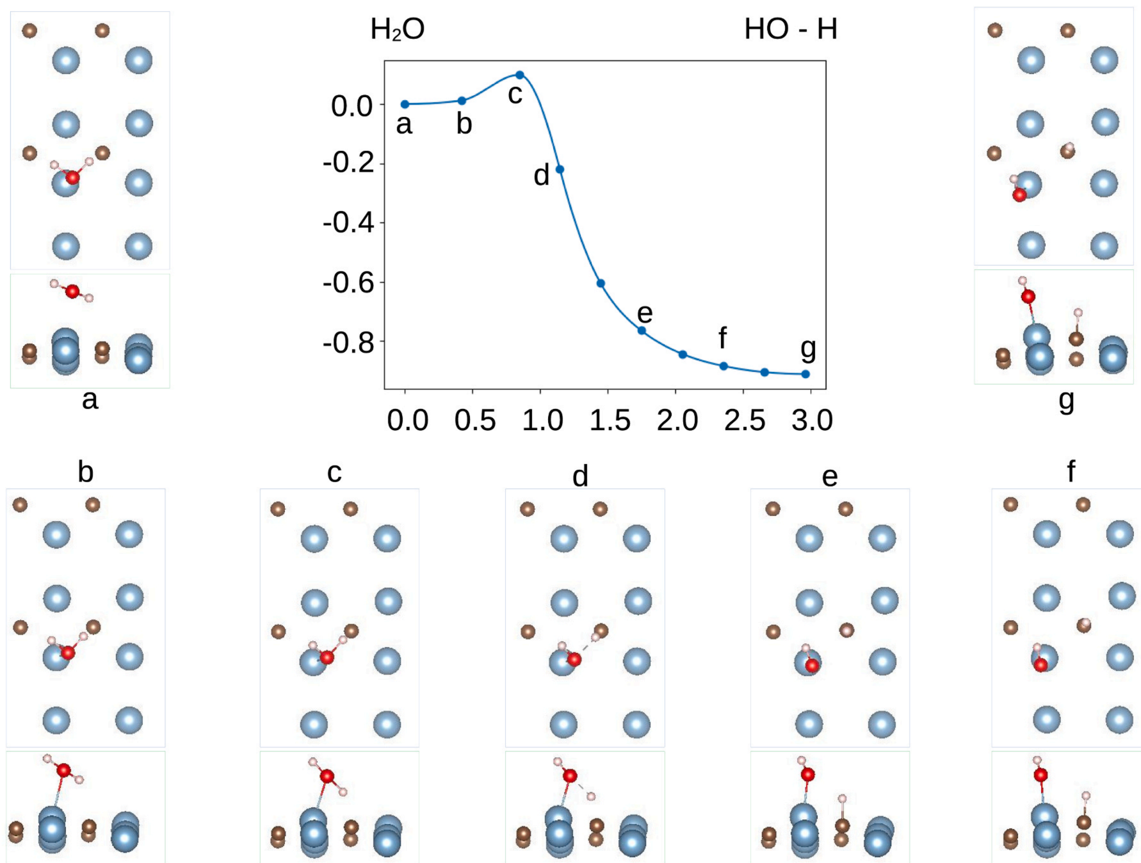

Fig. 5. MEP (central panel), energy (eV) and reaction pathway (Å) in vertical and horizontal axes, respectively. Top and side views of initial (a), five intermediate (b, c, d, e, f) and final (g) configurations for H_2O partial dissociation on the Al_2C nanosheet after ciNEB calculations, PBE results.

few meV in the cases of the dissociation of the other three molecules (H_2S, H_2Se and H_2Te). Fig. 6 shows, schematically, the MEP and some of the configurations toward the H_2S partial dissociation on the Al_2C nanosheet. The calculated activation energy is 15 meV. Fig. 7 shows, schematically, the MEPs and the configuration of the calculated maximum toward the H_2Se and H_2Te partial dissociations on the Al_2C nanosheet. As can be seen, wide maxima are found in these two cases as well as a small activation energies, 21 and 67 meV for H_2Se and H_2Te, respectively. It is, as well, worth mentioning that the activation energy to H_2X formation from the fragments HX-H, is larger than 1.5 eV on the Al_2C nanosheet (X = S, Se and Te). This could do the clean Al_2C nanosheet recovery kinetically difficult.

Molecular and dissociative adsorption, density of states, energy gaps. It is known that the PBE functional underestimates the energy gap between the valence and conduction bands for semiconductors. However, it was found that the HSE06 hybrid functional [22] shows an accurate description of the electronic properties of 2D semiconductors [23,24]. In this work, the HSE06 hybrid functional is used to calculate the density of states (DOS) and the energy gap (above-mentioned in section 2). Present research unveils the energy gaps for the systems H_2X molecule adsorbed on the Al_2C nanosheet for X = O, S, Se and Te, in the low coverage regime. For molecular adsorption, in all four cases, the results of the calculations of the density of states (DOS), reveal an increase of the energy gap with respect to that one calculated for the pristine Al_2C monolayer. The calculated energy gap between the valence and conduction bands is 1.078 eV, for the pristine Al_2C nanolayer, in good agreement with previous calculations [10,11,13]. This energy gap should be compared with 0.398 eV, that is the result of the PBE calculation. Fig. 8 displays that the calculated energy gaps between valence and conduction bands are 1.371, 1.338, 1.379 and 1.416 eV after H_2O, H_2S, H_2Se and H_2Te adsorption, respectively. These values represent an increase of ≈ 27.18%, ≈ 24.12%, ≈ 27.92% and ≈ 31.26%, respectively.

Figure 8 also reveals that the energy gaps between valence and conduction bands are calculated in 1.206, 1.363, 1.370 and 1.388 eV for partial dissociative adsorption as XH-H with X = O, S, Se and Te, respectively. These results represent an increase of ≈ 12.06%, ≈ 26.44%, ≈ 27.09% and ≈ 28.76%, respectively.

Comparisons between the energy gap obtained for the pristine Al_2C nanosheet (1.078 eV) and for the same nanosheet after molecular or dissociative adsorption of any of the four adsorbates, show that the Al_2C nanosheet is potentially useful o could be applied to measure the presence of any of these molecules in small quantities. As above-mentioned, the nanosheet energy gap increases strongly after molecular or dissociative adsorption of the studied molecules. The increase of the energy gap means the electrical conductivity decreases.

Furthermore, even though the differences in the calculated energy gaps between molecular and dissociative adsorption seem to be small, a comparison between both electrical conductivities could be calculated or estimated. The electrical conductivity, σ, could be estimated using equation (2)

$$\sigma = A\ exp\left(-\frac{E_g}{2K_BT}\right) \tag{2}$$

In equation (2) the terms A, K_B and T correspond to a pre-exponential factor, the Boltzmann constant (eV/K) and the absolute temperature (K), respectively. For any of the adsorbates, the rate between electrical conductivities could be

$$\frac{\sigma_{Al_2C-H_2X}}{\sigma_{Al_2C-HX-H}} = exp\left(\frac{-E_g(Al_2C-H_2X)+E_g(Al_2C-HX-H)}{2K_BT}\right) \tag{3}$$

In equation (3) the term E_g represent the corresponding energy gap

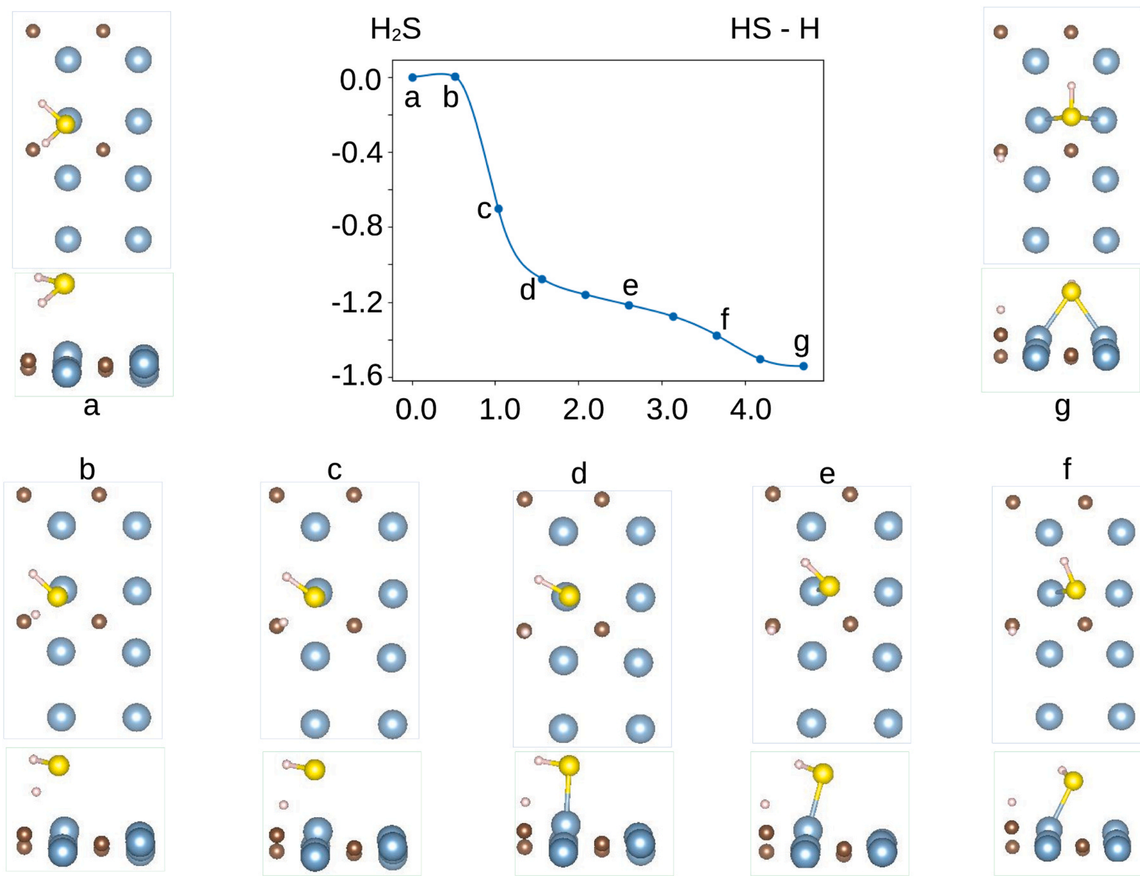

Fig. 6. MEP (central panel), energy (eV) and reaction coordinate (Å) in vertical and horizontal axes, respectively. Top and side views of initial (a), five intermediate (b, c, d, e, f) and final (g) configurations for H_2S partial dissociation on the Al_2C nanosheet after ciNEB calculations, PBE results.

whereas X stands for O, S, Se and Te. Using the calculated energy gaps, E_g, and T = 293 K, we find

$$\sigma_{Al_2C-HO-H} \approx 25.23\ \sigma_{Al_2C-H_2O} \tag{4}$$

$$\sigma_{Al_2C-H_2S} \approx 1.05\ \sigma_{Al_2C-HS-H} \tag{5}$$

$$\sigma_{Al_2C-HSe-H} \approx 1.20\ \sigma_{Al_2C-H_2Se} \tag{6}$$

$$\sigma_{Al_2C-HTe-H} \approx 1.74\ \sigma_{Al_2C-H_2Te} \tag{7}$$

Equation (4) shows the electrical conductivity of the Al_2C nanosheet in which the H_2O molecule is dissociated, HO-H, is more that 25 times larger than the nanosheet electrical conductivity when the adsorption is molecular. Equation (5) shows the only case, of the studied cases, in which the electrical conductivity of the Al_2C nanosheet with molecular adsorption, H_2S, is larger than the nanosheet electrical conductivity when dissociative adsorption takes place, HS-H adsorbed on the Al_2C nanosheet. Equation (6) shows the electrical conductivity of the Al_2C nanosheet in which the H_2Se molecule is dissociated, HSe-H, is about 20% larger than the nanosheet electrical conductivity for molecular adsorption, H_2Se adsorbed on the Al_2C nanosheet. As in the previous case, equation (7) displays the electrical conductivity of the system Al_2C nanosheet-dissociative adsorption HTe-H, is around 1.74 times the electrical conductivity of the system Al_2C nanosheet-molecular adsorption, H_2Te adsorbed on the Al_2C nanosheet. Following equations (4) to (7), assuming equation (2) can be applied, the Al_2C nanosheet seems to be a good or promising candidate to differentiate between molecular and dissociative adsorption in these four cases.

Molecular adsorption, charge analysis. Bader charge analysis applied on the isolated molecules, shows the oxygen atom takes ≈ 0.5904 *e* from every hydrogen atom whereas the hydrogen atoms take charge from the central atom in the other three studied molecules. Table 2 displays that every H atom takes ≈ 0.0114 *e*, ≈ 0.0552 *e* and ≈ 0.1622 *e* from the S, Se and Te central atoms. This is, as the size of the central atom is bigger, larger is the decrease of the negative charge on it after adsorption.

Upon adsorption, the negative charge on all four adsorbates increases slightly: 0.0891*e*, 0.0635*e*, 0.0449*e* and 0.0597*e* on H_2O, H_2S, H_2Se and H_2Te, respectively, see Table 2. This is, the H_2O molecule, as well as the other three molecules, are taking negative charge from the surface. This is very different from the H_2O molecular adsorption on metal surfaces, such as Al{100} and fcc{111} surfaces ([37,38] and references therein). On these metal surfaces, the H_2O molecule adsorbs on a metal atom via the $1b_1$ molecular orbital, mainly, and there is a decrease of the negative charge on the mentioned molecular orbital.

4. Conclusions

The interaction between small molecules, such as H_2O, H_2S, H_2Se and H_2Te, and the pristine Al_2C nanosheet has been studied using DFT and the London dispersion interactions. The optimized Al_2C nanosheet is *flat* whereas the lattice constants, a = 3.009 Å and b = 5.070 Å, and the C-C and Al-C bond lengths agree with previous studies.

Molecular interactions are weak even considering the London interactions. The range of adsorption energies is − 0.510 to − 0.158 eV for H_2O and H_2S, whereas these limits are − 0.588 to − 0.263 eV, respectively, when the London dispersion interactions are taken into account. In all adsorption configurations, the molecular plane is nearly parallel to the Al_2C nanosheet plane whereas the central atom is near on top of the Al atom.

Dissociative adsorption as XH-H, is strongly preferred to molecular interaction in all four cases. Following the PBE results, the larger the

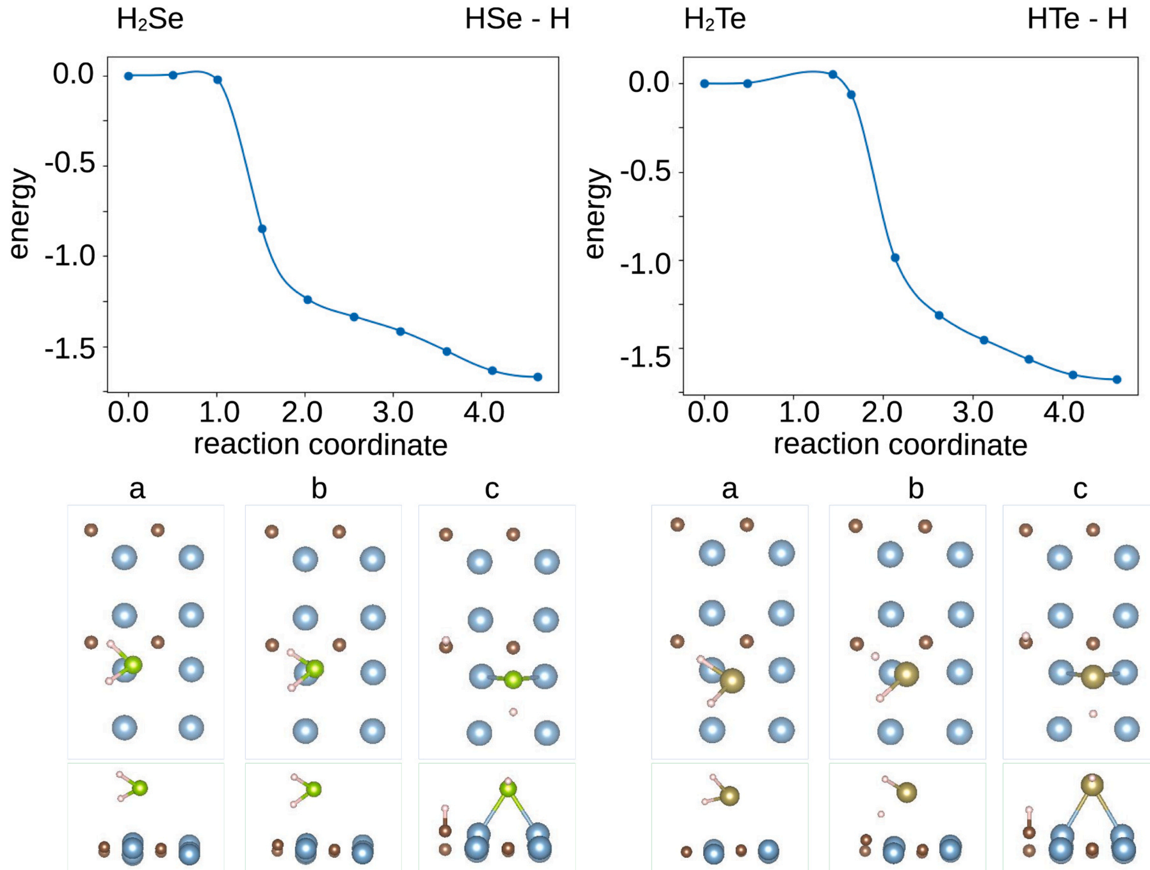

Fig. 7. MEPs (top panels), energy (eV) and reaction coordinate (Å) in vertical and horizontal axes, respectively. Top and side views of initial (a), intermediate (b) and final (c) configurations for H_2Se and H_2Te partial dissociation on the Al_2C nanosheet after ciNEB calculations, PBE results.

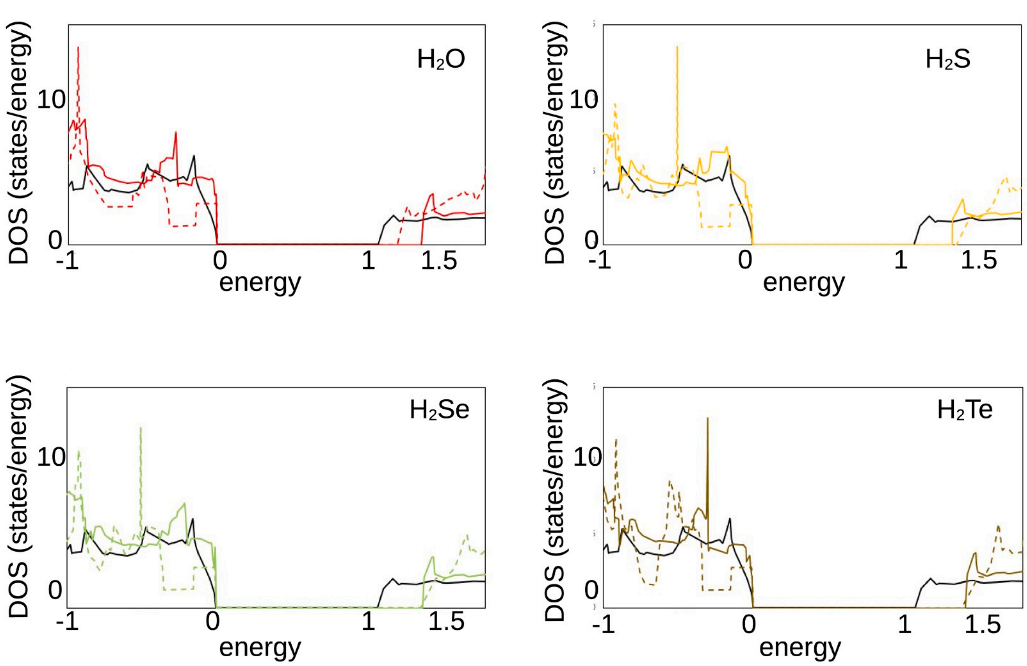

Fig. 8. DOS (states/energy) vs energy (eV), calculated using the HSE06 hybrid functional. Pristine Al_2C nanosheet in black continuous line. H_2X molecular (HX-H dissociative) adsorption for X = O, S, Se and Te in red, yellow, green, brown continuous (dashed) lines, respectively.

central atom is the stronger the interaction is, from − 1.422 to − 1.902 eV for HO-H and HTe-H, respectively. This trend is no longer true when the vdW interactions are taken into account. CiNEB calculations show the activation energy for H_2O dissociation to OH-H is only ≈ 0.1 eV, whereas it is just a few meV in the cases of H_2S, H_2Se and H_2Te dissociation.

Table 2
Bader charge calculated on the isolated adorbates, H_2X, and on the systems Al_2C-H_2X (in units of *e*).

atom	H_2X	Al_2C-H_2X	diff
O	7.1808	7.3751	0.1943
H	0.4096	0.3570	− 0.0526
H	0.4096	0.3570	− 0.0526
Total	8.0000	8.0891	0.0891
S	5.9772	6.1455	0.1683
H	1.0114	1.0021	− 0.0093
H	1.0114	0.9159	− 0.0956
Total	8.0000	8.0635	0.0635
Se	5.8896	5.9268	0.0372
H	1.0552	1.0466	− 0.0086
H	1.0552	1.0715	0.0163
Total	8.0000	8.0449	0.0449
Te	5.6756	5.6916	0.0159
H	1.1622	1.1659	0.0037
H	1.1622	1.2022	0.0401
Total	8.0000	8.0597	0.0597

Bader charge analysis shows that the negative charge on the Al_2C nanosheet decreases after molecular adsorption.

The calculated energy gap for the pristine Al_2C nanosheet is 1.078 eV. However, it is increased to 1.371, 1.338, 1.379 and 1.416 eV for molecular adsorption (H_2O, H_2S, H_2Se and H_2Te, respectively) and to 1.209, 1.363, 1.370 and 1.388 eV for partial dissociative adsorption. These results could be considered as an indication that the Al_2C nanosheet has a potential use as sensor of these molecules in molecular or dissociated state. However, experiments and experimental results are needed in order to reach definitive conclusions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work has been supported by Consejo de Investigaciones Científicas y Técnicas (CCT-CONICET-La Plata), Argentina and Departamento de Física, Facultad de Ciencias Exactas, Universidad Nacional de La Plata. (1900) La Plata, Argentina.

References

[1] R. Baughman, H. Eckhardt, M. Kertesz, Structure-property predictions for new planar forms of carbon: layered phases containing sp^2 and *sp* Atoms, J. Chem. Phys. 87 (1987) 6687.
[2] N. Narita, S. Nagai, S. Suzuki, K. Nakao, Optimized geometries and electronic structures of graphyne and its family, Phys. Rev. B 58 (1998) 11009.
[3] K.S. Novoselov, A.K. Geim, S. Morozov, D. Jiang, Y. Zhang, S. Dubonos, I. Grigorieva, A. Firsov, Electric field effect in atomically thin carbon films, Science 306 (2004) 666–669.
[4] S. Iijima, Helical microtubules of graphitic carbon, Nature 354 (1991) 56–58.
[5] S. Iijima, T. Ichihashi, Single-shell carbon nanotubes of 1-nm diameter, Nature 363 (1993) 603–605.
[6] H.W. Kroto, J.R. Heath, S.C. O'Brien, R.F. Curl, R.E. Smalley, C 60: Buckminsterfullerene, Nature 318 (6042) (1985) 162–163.
[7] H.J. Monkhorst, Activation energy for interconversion of enantiomers containing an asymmetric carbon atom without breaking bonds, Chem. Commun. 18 (1968) 1111.
[8] M.J.M. Pepper, I. Shavitt, P.V.R. Schleyer, M.N. Glukhovtsev, R. Janoschek, M. Quack, J. Comput. Chem. 16 (1995) 207.
[9] R. Hoffmann, R.W. Alder, C.F. Wilcox, Planar tetracoordinate carbon, J. Am. Chem. Soc. 92 (1970) 4992–4993.
[10] J. Dai, X. Wu, J. Yang, X.C. Zeng, Al_xC Monolayer Sheets: two-dimensional networks with planar tetracoordinate carbon and potential applications as donor materials in solar cell, J. Phys. Chem. Lett. 5 (2014) 2058–2065.
[11] Y. Li, Y. Liao, P. von RaguéSchleyer, Z. Chen, Al_2 Cmonolayer: the planar tetracoordinate carbon global minimum, Nanoscale 6 (2014) 10784.
[12] G. Jiménez-Cadena, J. Riu, F.X. Rius, Gas sensors based on nanostructured materials, Analyst 132 (2007) 1083.
[13] R. Rahimi, M. Solimannejad, A. Chaudhari, Toxic volatile organic compounds sensing by Al_2C monolayer: a first-principles outlook, J. Hazard. Mater. 403 (2021) 123600.
[14] R. Rahimi, M. Solimannejad, Sensing ability of 2D Al_2C monolayer toward toxic pnictogen hydrides: a first-principles perspective, Sens. Actuators A 331 (2021) 113000.
[15] G. Kresse, J. Hafner, Phys. Rev. B 49 (1994) 14251.
[16] G. Kresse, J. Furthmuller, Phys. Rev. B 54 (1996) 11169–11186.
[17] P. Blöchl, Phys. Rev. B 50 (1994) 17953.
[18] G. Kresse, J. Joubert, Phys. Rev. B 59 (1999) 1758.
[19] J.P. Perdew, J.A. Chevary, S.H. Vosko, K.A. Jackson, M.R. Pederson, D.J. Singh, C. Fiolhais, Phys. Rev. B 46 (1992) 6671–6687.
[20] J.P. Perdew, K. Burke, M. Ernzerhof, Phys. Rev. Lett. 77 (1996) 3865.
[21] J. Klimeš, A. Michaelides, Perspective: advances and challenges in treating van der waals dispersion forces indensity functional theory, J. Chem. Phys. 137 (2012) 120901.
[22] A.V. Krukau, O.A. Vydrov, A.F. Izmaylov, G.E. Scuseria, Influence of the exchange screening parameter on the performance of screened hybrid functionals, J. Chem. Phys. 125 (2006) 224106, https://doi.org/10.1063/1.2404663.
[23] M. Ghambarian, Z. Azizi, M. Ghashghaee, Remarkable improvement in phosgene detection with a defect-engineered phosphorene sensor: first-principles calculations, Phys. Chem. Chem. Phys. 22 (2020) 9677–9684.
[24] M. Ghashghaee, M. Ghambarian, Highly improved carbon dioxide sensitivity and selectivity of black phosphorene sensor by vacancy doping: a quantum chemical perspective, Int. J. Quantum Chem. 120 (16) (2020) e26265.
[25] G. Henkelman, H. Jónsson, A climbing image nudged elastic band method for finding saddle points and minimum energy paths, J. Chem. Phys. 113 (2000) 9901–9904.
[26] M. Dion, H. Rydberg, E. Schröder, D.C. Langreth, B.I.L. Van der Waals, Density functional for general geometries, Phys. Rev. Lett. 92 (2004) 246401.
[27] G. Román-Pérez, J.M. Soler, Efficient implementation of a vander Waals Density Funtional: application to double-wall carbon nanotubes, Phys. Rev. Lett. 103 (2009) 096102.
[28] J. Klimes, D.R. Bowler, A. Michaelides, Chemical accuracy forthe van der Waals density functional, J. Phys. Condens. Matter 22 (2010) 022201.
[29] K. Lee, E.D. Murray, L. Kong, B.I. Lundqvist, D.C. Langreth, Higher-accuracy van der Waals density functional, Phys. Rev. B 82 (2010) 081101(R).
[30] J. Klimes, D.R. Bowler, A. Michaelides, Van der Waals density functionals applied to solids, J. Phys. Rev. B 83 (2011) 195131.
[31] T. Thonhauser, V.R. Cooper, S. Li, A. Puzder, P. Hyldgaard, D.C. Langreth, Van der Waals density functional: self-consistent potential and the nature of the van der Waals bond, Phys. Rev. B 76 (2007) 125112.
[32] T. Kerber, M. Sierka, J. Sauer, Application of semiempirical long-range dispersion corrections to periodic systems in density functional theory, J. Comput. Chem. 29 (2008) 2088.
[33] E. Londero, E. Schröder, Role of van der Waals bonding in thelayered V_2O_5: first-principles density-functional calculations, Phys. Rev. B 82 (2010) 054116.
[34] G. Graziano, J. Klimes, F. Fernandez-Alonso, A. Michaelides, J. Phys. Condens. Matter 24 (2012) 424216.
[35] V.A. Ranea, P.L. Dammig Quiña, N.M. Yalet, General adsorption model for H_2S, H_2Se, H_2Te, NH_3, PH_3, AsH_3 and SbH_3 on the V_2O_5 (001) surface including the van der Waals interaction, Chem. Phys. Lett. 720 (2019) 58–63.
[36] V.A. Ranea, A DFT+U study of H_2O adsorption on the V_2O_5 (001)surface including van der Waals interactions, Chem. Phys. Lett. 730 (2019) 171–178.
[37] A. Michaelides, V.A. Ranea, P.L. de Andrés, D.A. King, First-principlesstudy of H_2O diffusion on a metal surface: H_2O on Al{100}, Phys. Rev. B 69 (2004) 075409.
[38] V.A. Ranea, Potential energy surface of H_2O on Al{111} and Rh{111} from theoretical methods, J. Chem. Phys. 137 (2012) 204702.
[39] V.A. Ranea, I. Carmichael, W.F. Schneider, DFT Investigation of intermediate steps in the hydrolysis of α-Al_2O_3 (0001), J. Phys. Chem. C. 113 (2009) 2149–2158.